Midterm Examination

Prof. J. Whitfield

Monday, July 15, 2024 Time allowed: 90 minutes Total: 60 points

Instructions. This exam consists of two long problems built around a single case. Read the case description carefully; later parts build on earlier ones, but each part can be attempted on its own using the information given. Point values are in brackets. Write clearly and explain your reasoning — the grading emphasizes correct strategic logic at least as much as the final numbers. You may use a simple calculator. No notes, books, or phones.

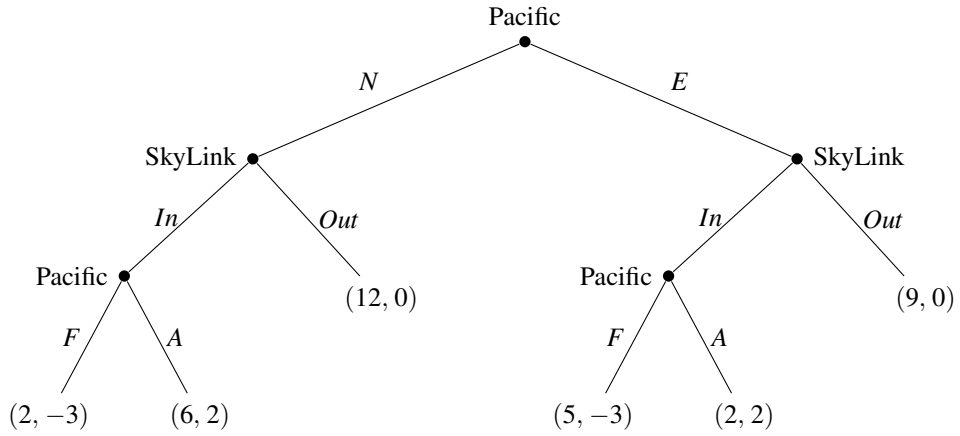
Name: _____ Student ID: _____

	Problem 1	Problem 2	Total
Points	32	28	60
Score			

The case: Pacific Air and SkyLink

Pacific Air is the only carrier flying a profitable regional route. A low-cost carrier, SkyLink, is considering entering the route. Before SkyLink decides, Pacific Air can choose to *expand* capacity by leasing additional aircraft (E), or *not* (N). Expansion costs 3 (in \$ millions, present value) and is sunk once incurred. SkyLink observes Pacific Air's choice and then decides whether to enter (*In*) or stay out (*Out*). If SkyLink enters, Pacific Air observes this and decides whether to start a fare war and *fight* (F) or *accommodate* (A).

Payoffs (Pacific Air, SkyLink), in \$ millions, are shown in the tree. The payoffs after E already subtract the expansion cost of 3. Expansion matters because idle aircraft make a fare war much cheaper for Pacific Air to wage.



Problem 1: Entry deterrence [32 points]

(a) How many pure strategies does each firm have in the game shown in the tree? List SkyLink's pure strategies explicitly. [4]

(b) Consider only the subgame that follows N. Write it in strategic (normal) form, with Pacific Air choosing between F and A and SkyLink choosing between *In* and *Out*. Find all pure-strategy Nash equilibria of this subgame. Which of them relies on a non-credible threat, and why? [6]

(c) Find the subgame perfect equilibrium of the whole game by backward induction. State each firm's complete strategy and the equilibrium outcome and payoffs. [8]

(d) Pacific Air's analysts want to know how robust the deterrence strategy is. Write the expansion cost as k instead of 3, so that the payoffs after E are $(12-k,\,0)$ after *Out*, $(8-k,\,-3)$ after F, and $(5-k,\,2)$ after A. [6]

(i) For which values of k does Pacific Air expand in the subgame perfect equilibrium? (Assume Pacific Air expands when indifferent.)

(ii) Now keep $k=3$ but suppose the gross fare-war payoff after expansion is F_g instead of 8 (so the net payoff after (E,In,F) is F_g-3). For which values of F_g does expansion deter entry? What does Pacific Air do in the SPE otherwise?

(e) Show that there is a Nash equilibrium of the whole game in which Pacific Air does *not* expand and SkyLink stays out. Explain why it is not subgame perfect. Pacific Air earns more in this Nash equilibrium than in the SPE; explain why, in the SPE, it nevertheless pays for capacity it never uses. [8]

Problem 2: The fare game after entry [28 points]

Two years later, regulators have awarded SkyLink landing slots on a different route that both airlines now serve. Every quarter, each airline simultaneously sets a *High* (*H*) or *Low* (*L*) fare. Quarterly profits (Pacific Air, SkyLink), in $ millions, are:

		SkyLink *H*	SkyLink *L*
Pacific Air	*H*	10, 6	2, 9
	L	14, 1	5, 3

Both airlines discount future profits by a factor $\delta \in (0,1)$ per quarter.

(a) Find the Nash equilibrium of the one-quarter game. Explain why this game is a prisoner's dilemma. [4]

(b) Suppose the route will be served for exactly 8 quarters, after which it closes (this is common knowledge). Find the subgame perfect equilibrium of the repeated game. Justify your answer. [4]

(c) Now suppose the route is served indefinitely. Consider the grim-trigger strategy profile: each airline sets H in the first quarter and continues to set H as long as both airlines have always set H; after any deviation, both set L forever. Find the smallest δ for which this profile is a subgame perfect equilibrium. Which airline's incentive constraint binds, and why does it make sense that it is this one? [8]

(d) Suppose that, in addition to discounting, each quarter there is a 20% chance that the route is permanently closed (independently across quarters), after which both airlines earn zero on it. [6]

(i) Find the smallest δ for which grim trigger sustains (H,H).

(ii) Suppose the airlines discount exponentially at a continuously compounded annual interest rate r, so $\delta = e^{-r/4}$. Express the condition in (i) as an upper bound on r, and decide whether cooperation is sustainable at $r = 10\%$.

(e) Pacific Air's CEO proposes a "forgiving" strategy instead of grim trigger: after any deviation, both airlines set L for exactly m quarters and then return to (H,H) (and any deviation during a punishment phase restarts it). Show that with $m = 1$ SkyLink would always deviate, whatever $\delta < 1$. For $m = 2$, find the smallest δ for which neither airline wants to deviate from (H,H). (You may take as given that nobody wants to deviate during a punishment phase.) [6]

— End of examination —